Pedireddla Sahil Sriram
Computer Science & Engineering
Indian Institute of Technology Bombay

22B1062
B.Tech.
Gender: Male
DOB: 11/02/2005

Examination	University	Institute	Year	CPI / % Credits
Graduation	IIT Bombay	IIT Bombay	2026	
Intermediate	BIEAP	Sri Chaitanya college	2022	
Matriculation	CBSE	Dr. KKR's Gowtham School	2020	

Scholastic Achievements

- Secured **All India Rank 332** in **Joint Entrance Examination-Advanced** among **150,000**+ candidates.(*2022*)
- Secured **99.951 percentile** and an **All India Rank 520** in **JEE-Main** among **1 million**+ candidates. (*2022*)
- Secured **736** rank in the **SX** stream in the prestigious **KVPY** examination conducted all over India (*2022*)
- Secured **State Rank 38** in **AP EAPCET** among **280K**+ candidates conducted by the **APSCHE** (*2022*)
- Secured **State Rank 70** in **TS EAMCET** among **150K**+ candidates conducted by the **TSCHE** (*2022*)

Internship Experience

Strategic Sales Route Intelligence for Market Penetration
May 2025 - July 2025
Summer Intern — ***JSW Paints***

- Implemented route optimization with **Google OR-Tools VRP solver** and priority constraints for efficient scheduling.
- Processed geospatial data using **Geopandas**(Python) and OpenRouteService API for accurate distance-based routing.
- Developed **Folium** and **Leaflet.js** visualizations with color-coded, zone-specific routes to improve field team navigation.

Key Projects

Cricket Score Prediction Model | *WiDS Project*
(December 2023 - January 2024)
Mentor : Siddharth, Artificial Intelligence and Machine Learning

- Developed a **Cricket score prediction model** using **Machine Learning** for highly accurate score forecasting.
- Performed **EDA** on cricket datasets by identifying patterns and **preprocessing data**, for robust model training.
- Integrated the model into an **end-to-end Streamlit app**, enabling real-time, user-friendly **score predictions**.

Social Rating and Review Platform | *Course Project*
(January 2025 - April 2025)
Prof. Sudarshan | Database and Information Systems

- Automated recommendations using collaborative filtering in **Node.js** and **PostgreSQL**, enhancing content discovery
- Streamlined **authorization, friends/follow, and suggestions** in **React and Express**, enhancing interaction.
- Optimized data handling with normalized schema and modular APIs in a **relational database**, improving scalability

Computer Networks | *Course Project*
(August 2024 - November 2024)
Prof. Vinay J Ribeiro | Computer Networks

- Designed **PHY** and **MAC** layers using audio based transmission with encoding, error correction, CRC codes, and collision handling protocols like **CSMA/CA** for reliable **multi-node communication** without physical medium.
- Automated network analysis using **TCP and UDP** scripts for bandwidth estimation and throughput evaluation.
- Developed network routing with TTL-based tables and broadcasts improving path discovery and route updates.

Exploring Operating Systems | *Course Project*
(January 2024 - April 2024)
Prof. Mythili Vutukuru | Operating Systems

- Explored **xv6 OS** internals, implemented **reader-writer locks** using **pthreads** with reader & writer priority modes.
- Built a basic **filesystem** in the OS environment with **file management**, **storage allocation**, low-level **system calls**.
- Implemented **IPC** using **Unix domain sockets, POSIX shared memory**, pipes for data exchange between processes.

Cache Optimization | *Course Project*
(August 2023 - November 2023)
Prof. Biswabandan Panda | Digital Logic Design and Computer Architecture Lab

- Performed simulation and comparative evaluation of **Cache replacement policies**, including **LRU, LFU, FIFO,** and **BIP** within the **ChampSim environment** across a wide array of diverse workloads using numerous trace files
- Designed a **Stream prefetcher** to analyze memory access patterns and implement **effective prefetching strategies**

GCC Like Compiler | *Course Project*
(January 2025 - April 2025)
Prof. Uday Khedkar | Compilers

- **Engineered a compiler** for a subset of the C language (**SCLP**) using Flex & Bison, incorporating robust type-checking
- Generated IRs like **Three Address Code (TAC)** and **Register Transfer Language (RTL)** for code optimisation.
- Translated RTL statements into **assembly code**, managed stack frames, along with function **prologues** and **epilogues**.

Other Projects

Machine Learning | *Course Projects* *(January 2024 - April 2024)*
Prof. Swaprava Nath | Artificial Intelligence and Machine Learning
- Built CNNs with **Convolution**, **ReLU**, MaxPool, FC & **Softmax** layers on the **MNIST** dataset for digit recognition.
- Implemented **forward pass**, **backpropagation**, and **PCA** using PyTorch & NumPy with optimized time complexity.
- Developed AI for **Tic Tac Toe & Notakto** using **backward induction**, **Minimax**, and **alpha-beta pruning**.

Algorithmic Trader | *Course Project* *(December 2023 - January 2024)*
Prof. Ashutosh Gupta | Data Structures and Algorithms
- Engineered an autonomous **algorithmic trading system** utilizing **data structures and algorithmic** strategies
- Developed various effective **trading algorithms** targeting **buying low, selling high** strategies in financial markets
- Applied **recursive method** to interpret market data aimed at **maximizing** trading outcomes while ensuring fairness

Auto-Completion of Sentences | *Course Project* *(August 2023 - November 2023)*
Prof. Ashutosh Gupta | Data Structures and Algorithms
- Implemented a code that autocompletes sentences by using a **trie** data structure and **KMP searching algorithm**
- Generated all the possible auto-complete sentences and the **most probable** auto-complete suggestion based on the **repetition** of proposed auto-complete word and the preceding word in the query appeared in the user's text history

File Compression | *Course Project* *(August 2023 - November 2023)*
Prof. Ashutosh Kumar Gupta | Data Structures and Algorithms
- Implemented **Huffman coding**, **Run-Length Encoding (RLE)**, and gamma parameter for efficient data compression.
- Combined **LZ77** with Huffman coding to design a compression scheme resembling the **DEFLATE algorithm**.

Query Processing in MIPS | *Course Project* *(August 2023 - November 2023)*
Prof. Biswabandan Panda | Digital Logic Design and Computer Architecture
- Built **vector** and **heap** data structures in **x86 assembly language** employing a custom header-based framework.
- Implemented **merge sort** and **binary search** procedures in **MIPS assembly** for time-efficient query processing.

Minesweeper-like Cricket | *Course Project* *(March 2023 - May 2023)*
Prof. Kameswari Chebrolu | Software Systems Lab
- Developed a cricket-themed grid game using **HTML, CSS, and JavaScript**, for gameplay & interactive UI.
- Enhanced visuals with **CSS** effects and added multiplayer and power-up features via **JavaScript** for engaging **UX**.

Line Follower Bot | *Course Project* *(March 2023 - May 2023)*
Prof. Ankit Jain, Prof. Joseph John | Makerspace
- Built an **Arduino-based** white line follower bot with line-following and dumping algorithms for 300g payload delivery.
- Tested the bot's navigation and dumping mechanisms on varied and inclined surfaces for performance verification.

RAG Chatbot using Ollama 3 and LangChain | *Self Project* *(July 2025)*
- Built a local RAG Chatbot by integrating **Ollama 3** for inference with **LangChain**-based document embeddings.
- Designed an interactive **Streamlit** interface that supports file upload and enables conversational querying.

Technical Skills

Programming	C, C++, Python, Java, Bash, Lex, Yacc ,VHDL, Assembly, Prolog, Haskel
Development	HTML, CSS, JavaScript, PHP, Express, Node.js, PostgreSQL, MySQL
Libraries	Matpltlib Numpy Scikit-learn Pandas Pytorch
Office Tools	LaTeX, git, Docker, Sed, Awk, Wireshark

Courses Undertaken

Computer Science:	Computer Programming and Utilization, Software Systems Lab, Data Structures and Algorithms, Discrete Structures, Data Analysis and Interpretation, Digital Logic Design and Computer Architecture, Design and Analysis of Algorithms, Artificial Intelligence and Machine Learning, Automata Theory and Logic, Operating Systems, Abstractions and Pardigms for Programming, Computer Networks, Database and Information Systems, Compilers
Mathematics:	Calculus, Linear Algebra, Differential Equations.
Others:	Classical Physics and Theory of Relativity, Quantum Physics, Makerspace, Physical Chemistry, Organic and Inorganic Chemistry,Biology.

Extracurricular Activities

- **National Cadet Corps(NCC) :** Completed one year of military training at **NCC, IIT Bombay** (*2023*)
- Choreographed and participated in a Dance Performance in a Traditional Day event hosted by the CSE Dept. (*2025*)